



SIMATS ENGINEERING

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Student Name: A.Rakesh

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Course Code: CSA0504

Slot:A

Course Name: DATA BASE MANAGEMENT SYSTEM

Course Faculty: DR.R.KESAVAN

Project Title: National Vocational Certification and Exam Slot-Booking System:An Integrated Study in File Organization, Indexing, Hashing, Query Optimization, and Transaction Management

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Candidate Registration and Certification Records
Module 1 - Indexed Candidate Master File

New Candidate Registration

Full Name: Trade/Vocation:

Aadhaar/Other ID: Certification Level:

Candidate Master Index (Hashed Bucket View)

Hash Bucket	Candidate ID	Name	Status
3-424	CND-2281	Ravi Kumar	Certified
3-424	CND-2282	Srinivasan	Pending
3-427	CND-3240	Ajith Das	Certified
3-432	CND-2381	Priya Vaseen	Certified

Number Records in Certification and Exam Slot-Booking System - 11 Modules

Project Description:

This module manages the onboarding of vocational candidates into the certification system and maintains their certification status as a persistent, efficiently searchable master file, forming the foundational data layer that the exam booking and result-tracking modules depend on. It captures candidate demographic and vocational details such as name, trade or vocation applied for, contact information, and identity proof through a registration interface, assigns each candidate a unique Candidate_ID at the time of insertion, and tracks the certification lifecycle status — Pending, Under Review, Certified, or Rejected — for each trade, while also supporting lookup, update, and status-change operations on individual records.

From a database design standpoint, the module applies hashing by generating a hash key from a candidate's identity number, such as an Aadhaar or national ID, to place and retrieve records in bucket storage, giving near constant-time average lookup instead of a linear scan; it is complemented by indexed sequential access or B+ tree indexing on Candidate_ID to support fast range queries, such as listing all candidates registered within a given month, and ordered traversal of the file..

Student Signature:

Guide Signature